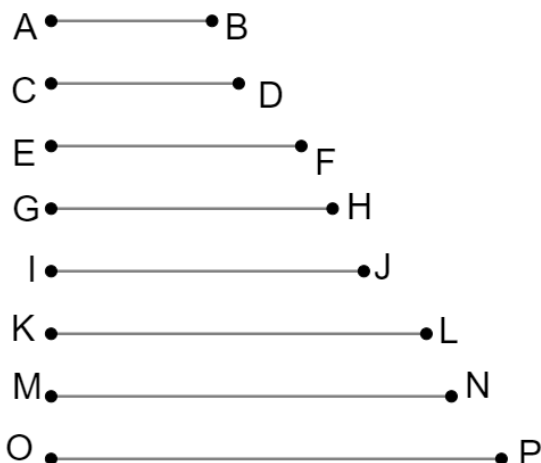
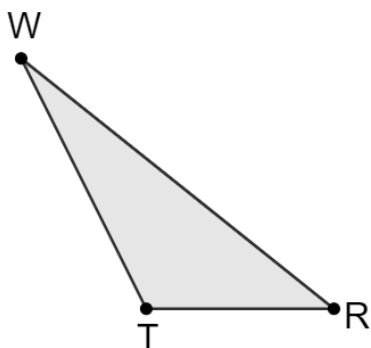


Geometry
Unit 5
Lesson 1 Practice

Name _____
Date _____
Period _____

Triangle WRT and 8 line segments are shown.

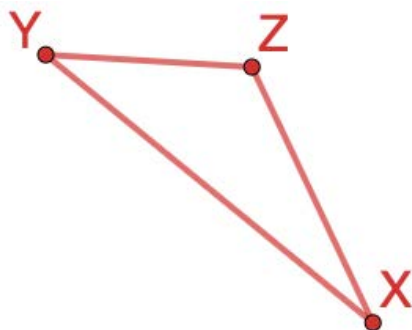


Use patty paper, a ruler, or edge of a piece of paper to determine which line segments have the same measure as each side of triangle WRT .

Complete each.

1. $WT = \underline{\text{GH}}$
2. $RT = \underline{\text{CD}}$
3. $WR = \underline{\text{MN}}$
4. Construct a triangle using the line segments that have the same measure as each side of triangle WRT . Name the new triangle XYZ .

Answers may vary.



5. Complete the statement.

The three sides of triangle WRT are congruent to the three sides of

triangle XYZ because $WT =$

$WR =$

- ☐ XY .
- ☐ YZ .
- ☐ XZ .

- ☐ XY ,
- ☐ YZ ,
- ☐ XZ ,

$RT =$

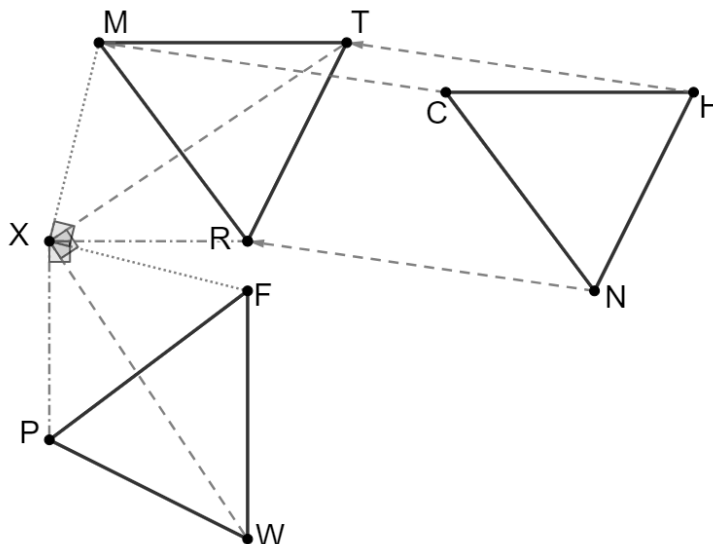
- ☐ XY ,
- ☐ YZ ,
- ☐ XZ ,

and

6. Determine if the angles of $\triangle WRT$ and $\triangle XYZ$ are congruent. Explain how you determined your answer.

$\triangle WRT \cong \triangle XYZ$ by the Side-Side-Side Congruence Postulate because the three sides of $\triangle WRT$ are congruent to the three sides of $\triangle XYZ$.

A series of transformations were applied to $\overline{CH}$, $\overline{HN}$, and $\overline{NC}$.



Complete the statements.

7. The first transformation applied to $\overline{CH}$, $\overline{HN}$, and $\overline{NC}$ is a translation. This results in $CH =$ MT , $HN =$ TR and $NC =$ RM .
8. Therefore, by the definition of congruence, $\triangle HNC \cong \triangle TRM$.
9. The second transformation applied to $\overline{MR}$, $\overline{RT}$, and $\overline{TM}$ is a rotation. This results in $MR =$ FP , $RT =$ PW and $TM =$ FW .
10. Therefore, by the definition of congruence, $\triangle TRM \cong \triangle$ WPF .